SUBJECT	Lower Secondary Science Year 8
TOPIC	Ecosystem Processes
ASSIGNMENT	Four
STUDENT NAME	

FEEDBACK AND REFLECTION:

- ☐ I have read the feedback on my last assignment carefully.
- ☐ I have considered whether any of that feedback is relevant to this assignment.

One thing I will try to do either the same or differently in this assignment as a result of my previous feedback is: ____

____

____

____

ASSIGNMENT GUIDANCE:

- You should complete this assignment by carrying out Practical 2: Population on this sheet and then handwriting in black ink both the Practical 2: Population questions and the assignment questions that follow on, unless you have obtained specific permission to submit typed work. Then scan it in and submit it, as a single document.
- This is not a timed assignment. We estimate that this assignment may take you between 1 and 2 hours for the practical, and between 40 and 60 minutes for the assignment itself. However, if you need to take longer, then please do so.
- Make sure you write your full name clearly in the box at the top of this page.
- In order to complete this assignment, you will need a calculator, ruler, your coursebook and notes.

LS Science Assignment Four

PRACTICAL 2: POPULATION

How does the population of daisies differ with different environmental conditions?

When scientists want to work out the population of an area, rather than (for example) counting every single daisy in a 100m² field, they will use a quadrat.

A quadrat is a square of metal, usually 1m² in size.

They will place it at **random** points in the field and count all the daisies they can see inside that square.

If they do this ten times, they will have the number of daisies in 10m².

If they multiply this number by 10, they will have an *estimate* of how many daisies are in the whole 100m² field!

Now, you may not have daisies in your area, and that is okay. It also does not have to be that big. You could do this experiment in your garden, or scale it down, make a really small quadrat, and sample the lichen on two different trees! Look around and decide what you want to investigate …

Apparatus

- Tape measure.
- Quadrat – you can make this at home out of cardboard. Just measure out the size you want, cut it out (ask for help please!) and trim it - make sure you have got a good chunky frame of cardboard of any size around that square hole.
- Clipboard and pen to take notes while you are sampling.

1. This question covers the practical.

Sampling

a. Why do you think they have to sample **randomly**? Why can't they just choose the spots with the most daisies and test those?

____

(1 mark)

Variables

b. What is the organism you are going to sample? Their population is now your dependent variable.

____

(1 mark)

c. What is your independent variable going to be? E.g. shade vs sun, wet vs dry, tree in the garden vs tree in the street?

____

(1 mark)

Method

d. This will be different for every student because you will all have access to different sites. So you are going to write your method today. The information at the top of the page will help you, but also think about the following things:

- How big is your sample area? How did you measure it and make sure the two sites were the same?
- How big was your quadrat? How many quadrats would it take to fill up your whole sample area?
- How many quadrats did you sample?
- How did you try to make sure your placement was random?

____

____

____

____

____

____

____

(4 marks)

Results

e. Put your results into the table below.

Quadrat number	Population of organism in each quadrat.	
	Site 1	Site 2

(1 mark)

f. Now can you work out what the population is of each area?

____

____

(2 marks)
(Total 10 marks)

ECOSYSTEM PROCESSES

2.

a. Complete the reaction representing photosynthesis below

Water + ____________________ ⟶ glucose (starch) + ______________

(2 marks)

b. How do plants obtain the energy for this process? Name the pigment involved.

Energy= _____________________ Pigment = ___________________

(2 marks)

c. Outline the role of stomata in this process

____

____

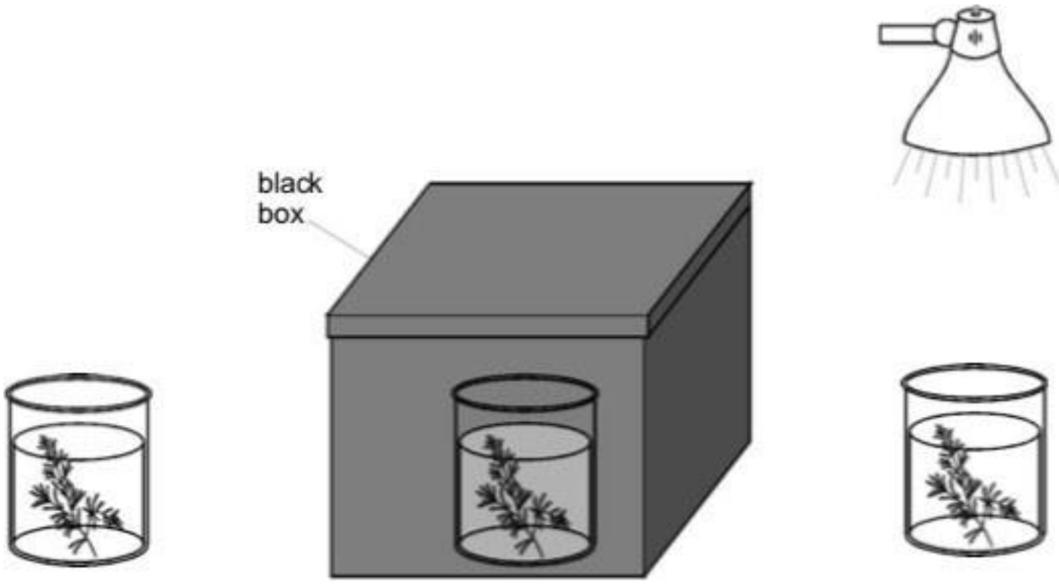
____

(1 mark)

(Total 5 marks)

3.

a. Alex poured some pond water into three beakers. She then put waterweed into each beaker. She put the beakers in different places.



Beaker A in normal daylight

Beaker B in the dark

Beaker C in bright light

i. In which beaker would the waterweed have the highest rate of photosynthesis? Give the correct letter.

____

(1 mark)

ii. The waterweed in the box changed from dark green to pale yellow. Why did this happen?

____

(1 mark)

b. In the school pond there were lots of water lilies with large leaves covering the surface. There were **not** many plants growing below the surface. Suggest a reason for this.

____

____

(1 mark)

c. In another experiment, Alex put similar pieces of waterweed into two more beakers of pond water. She added fertiliser to one of them. She kept them both by a window.

beaker D containing waterweed and pond water

beaker E containing waterweed and pond water plus fertiliser

i. Alex added fertiliser to beaker E. Suggest the results of this experiment.

____

____

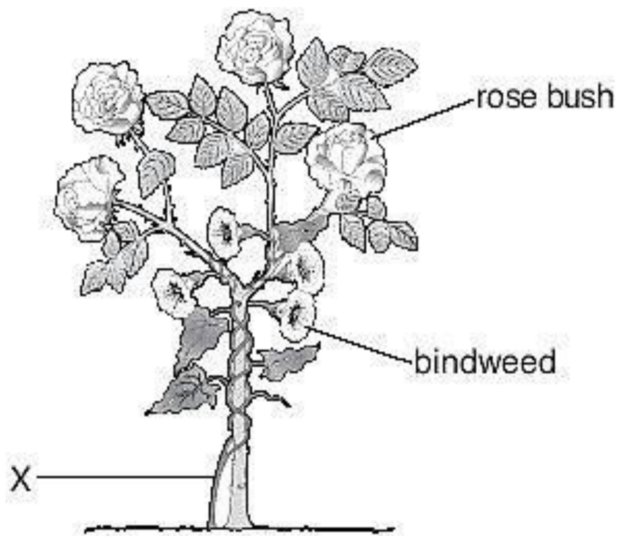
(1 mark)

ii. What do fertilisers contain to help plants grow? Tick the correct box.

fat	☐
sand	☐
minerals	☐
sugar	☐

(1 mark)
(Total 5 marks)

4. Bindweed is a plant that grows tightly around other plants. The drawing below shows bindweed growing around a rose bush.



a. Complete the sentences below. Choose from the words in the list.

air **light** **support** **water** **minerals**

i. Bindweed grows as high as possible on the rose bush so that the bindweed can get as much __________________as possible.

(1 mark)

ii. Bindweed grows around the rose bush because the rose bush provides _______________for the bindweed.

(1 mark)

b. A gardener cut through the stem of the bindweed at X. Two days later the bindweed above X was dead.

Why did the bindweed die? Tick the correct box.

no air	
no warmth	
no light	
no water	

(1 mark)

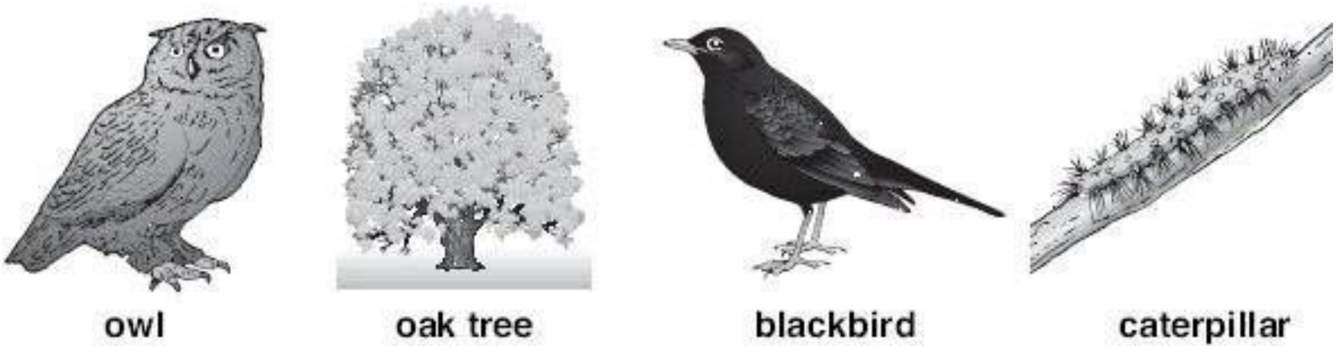
c. The gardener adds fertiliser to the soil to help her rose bushes to grow well. What do plants get from the fertiliser? Tick the correct box.

acids	
sugars	
minerals	
vitamins	

(1 mark)

(Total 4 marks)

5. The drawings below show four living things found in a wood.



not to scale

- Caterpillars eat oak leaves.
- Owls eat blackbirds.
- Blackbirds eat caterpillars.

a.

i. Complete the food chain for these four living things.

oak tree → ________________ → ________________ → ________________

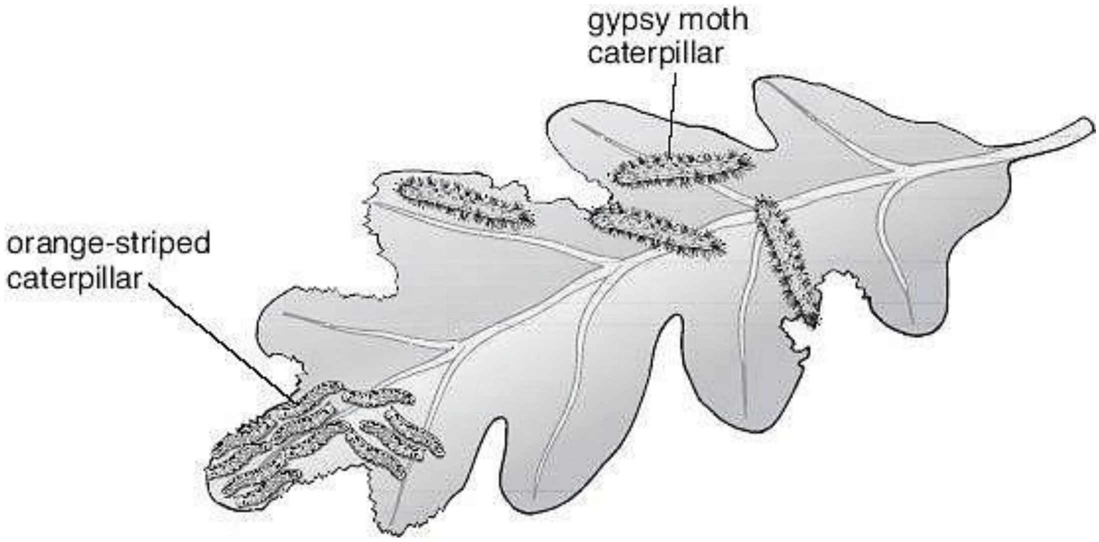
(1 mark)

ii. Why is an oak tree called a producer? Tick the correct box.

it loses its leaves in autumn	
it makes food by photosynthesis	
its flowers are tiny	
its leaves will **not** rot	

(1 mark)

b. On one oak tree, there were two types of caterpillar.



All the caterpillars were eating the leaves. The number of gypsy moth caterpillars increased.

i. What happened to the number of orange-striped caterpillars?

____

(1 mark)

ii. Explain your answer.

____

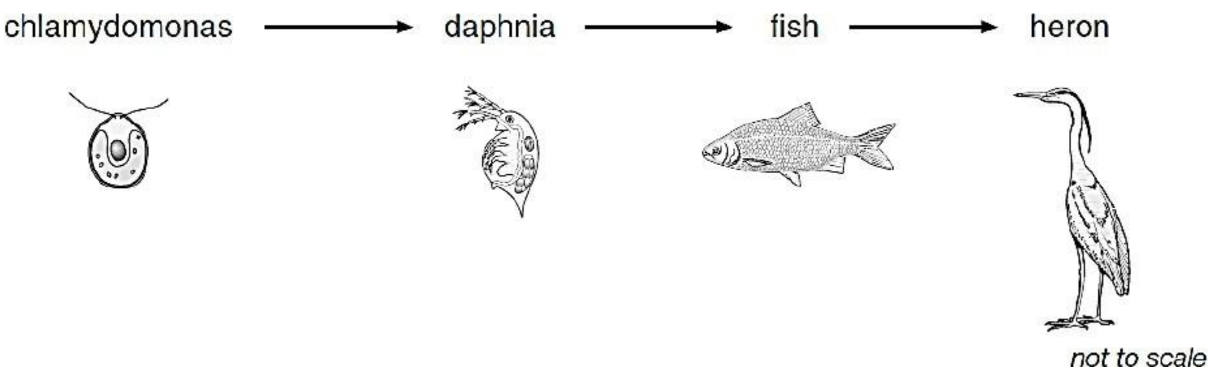
____

(1 mark)

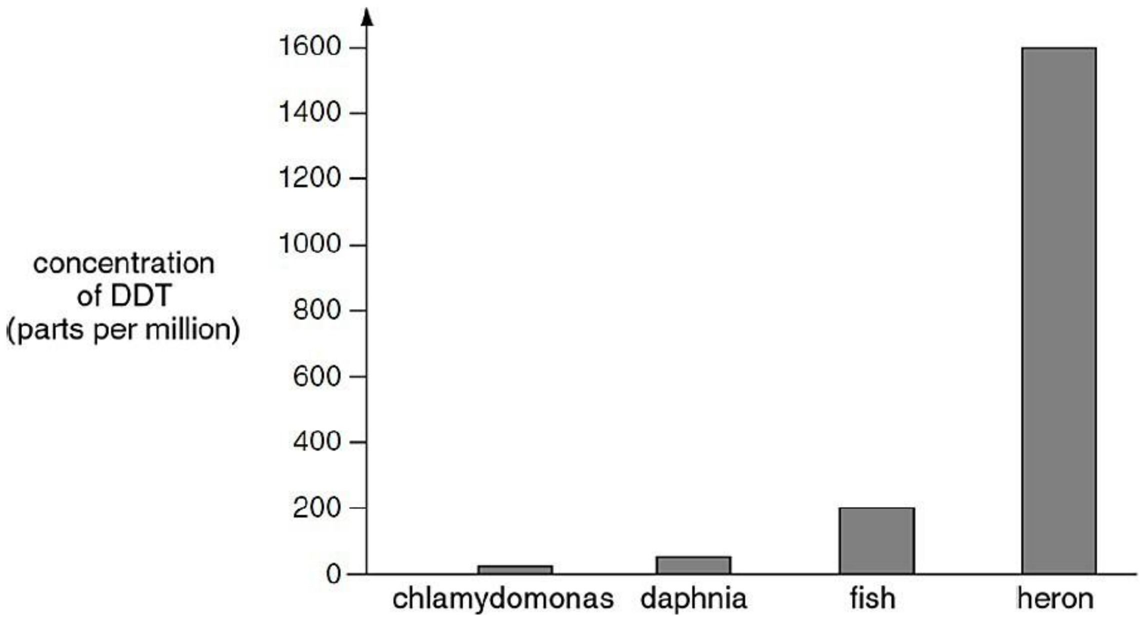
(Total 4 marks)

6. Scientists measured the concentration of the insecticide, DDT, in three animals and microscopic plant called chlamydomonas.

   a. The food chain for these four organisms is shown below.



The bar chart shows the concentration of DDT in the four organisms.



Give **one** reason for the difference in the concentration of DDT in these organisms.

____

____

(1 mark)

b. In 1970 the average concentration of DDT in the tissues of sea lions in California was 760 parts per million. Nearly half the sea lion pups born that year died because of high levels of DDT in their tissues.

How does DDT get from the body of a mother sea lion into the body of her pup:

i. **before** the pup is born?

____

____

(1 mark)

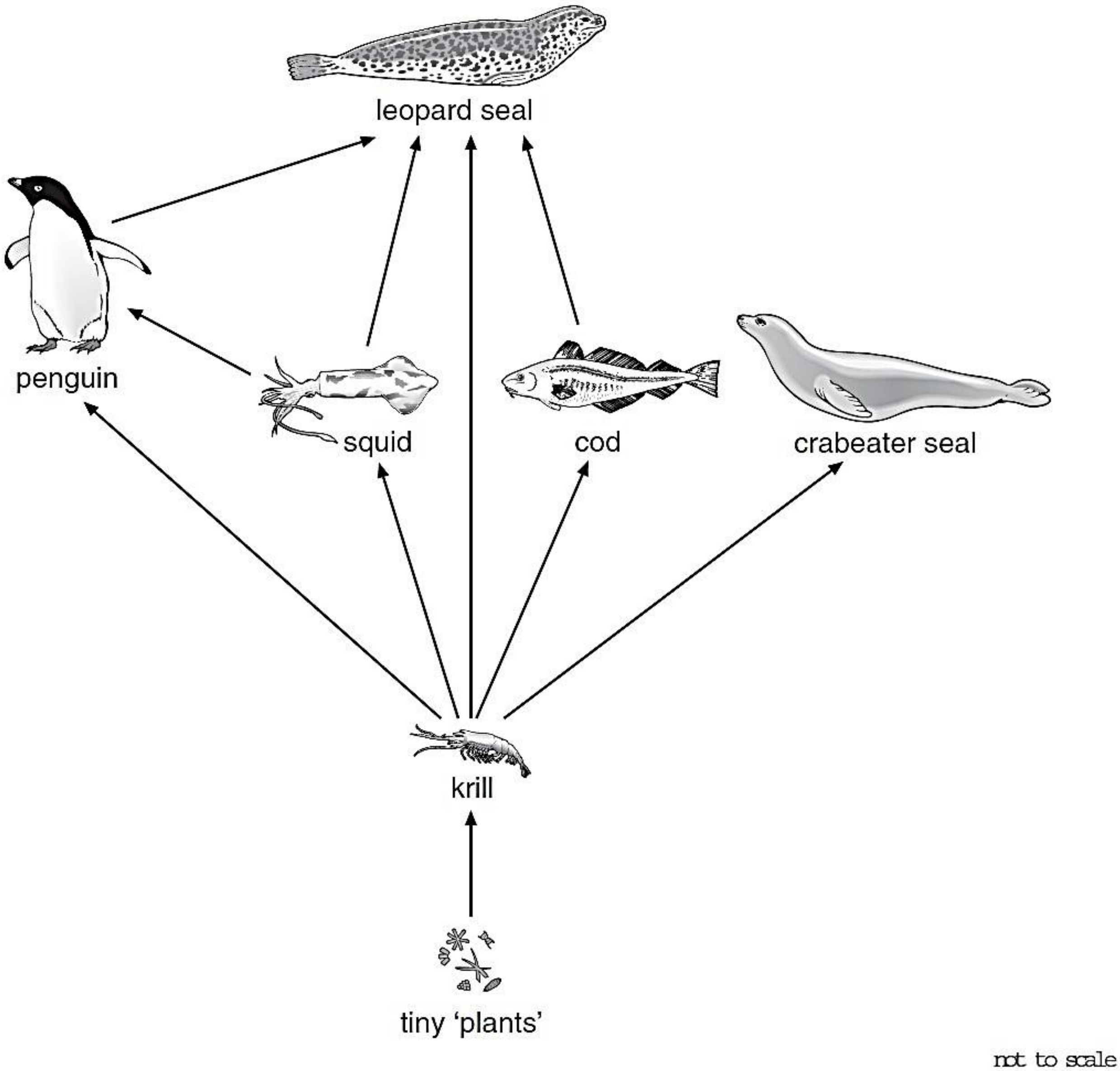
ii. **after** the pup is born?

____

____

(1 mark)
(Total 3 marks)

7. The drawing below shows part of a food web in the sea around Antarctica.



a. From the food web give the name of two animals that eat only krill.

1. ____________________

2. ____________________

(1 mark)

b.

i. Which word describes the plants in a food web? Tick the correct box.

producers	☐
predators	☐
herbivores	☐
carnivores	☐

(1 mark)

ii. Krill are small animals that eat tiny plants. Which word describes krill in the food web? Tick the correct box.

producers	☐
predators	☐
herbivores	☐
carnivores	☐

(1 mark)

c.

i. Crabeater seals eat krill. Fishermen catch large amounts of krill from the sea. How would a decrease in the number of krill affect the **number** of crabeater seals?

____

____

(1 mark)

ii. Look at the food web. Leopard seals also eat krill.

A decrease in the number of krill would affect the crabeater seals sooner than it affects leopard seals. Give the reason for this.

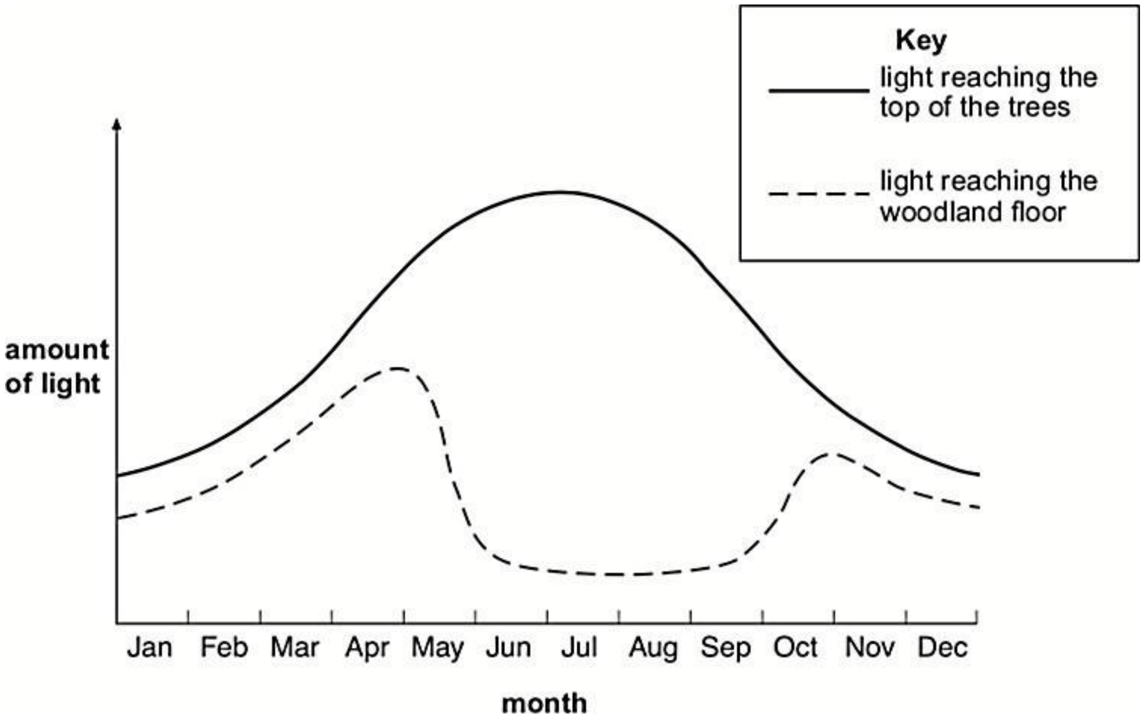
____

____

(1 mark)
(Total 5 marks)

8. The drawings below show the trees in a woodland area at the beginning of May and at the end of May.

The graph below shows the amount of light reaching the top of the trees and the woodland floor over one year.



a. Why does the amount of light reaching the woodland floor decrease during May?

____

(1 mark)

b. Plants grow on the woodland floor. Explain why these plants grow bigger **and** faster when there is plenty of light.

____

____

(2 mark)

c. Respiration takes place in the cells of all plants. Complete the word equation for respiration.

oxygen **+** ________________ ⟶ carbon dioxide **+** ___________________

(2 marks)
(Total 5 marks)

TOTAL FOR ASSIGNMENT 41 MARKS